FitFindr goes live — and keeps a record

Capstone, Shape A: deploy + instrument (run locally with logs)

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What changed: script → logged server

BEFORE

- Ran only as python app.py ask
- Exited after one answer
- No record of what happened
- Cost and speed were guesses

AFTER

- serve.py stays up: POST /ask
- One JSON line per request in logs/serve_log.jsonl
- Logs time, status, outcome, tokens
- capstone_trials.py: 11 requests per trial

How to run it: three commands

1

Install once

```
pip install -r requirements.txt    (key goes in .env)
```

2

Start the server

```
$env:AI201_CACHE="0"; python serve.py
```

3

Run a trial

```
python capstone_trials.py --label before --trial 1
```

The error the log caught: two bad inputs crash

Bad input sent	Response	Logged?
{} (empty object)	400	yes
blank query	400	yes
text, not JSON	400	yes
["vintage tee"] (a list)	500 crash	NO
{"query": 123}	500 crash	NO

Why

serve.py calls .get() and .strip() without checking the type first. A list has no .get(); a number has no .strip(). Flask throws, and my log line is never reached.

Fix: check the type up front and return 400, so it is logged too.

Deploy failed for AI201-project 2-fitfindr

deploy failed for ai201-project2-fitfindr-starter-v2026 Inbox ×

Render
We encountered an error during the deploy process for ai201-project2-fitfindr-starter-v2026. This means your deploy didn't complete successfully and your latest

Render <no-reply@render.com>
to me



We encountered an error during the deploy process for ai201-project2-fitfindr-starter-v2026. This means your deploy didn't complete successfully and your latest changes may not be live.

- **Commit:** serve_log.jsonl logs update and capstan_before+trial1.md result add

View Logs

Output capstone_trials.py – label before – trial 1

```
python capstone_trials.py --label before --trial 1

  [200]   7.6s  matching   vintage graphic tee under $30
  [200]   4.7s  matching   denim jacket under $50
  [200]   4.5s  matching   90s track jacket in size M
  [200]   5.0s  matching   silk slip dress in midi length under $40
  [200]   4.4s  matching   platform sneakers size 8
  [200]   2.8s  impossible designer ballgown size XXS under $5
  [400]   2.0s  bad input  empty JSON object
  [400]   2.0s  bad input  blank query
  [400]   2.1s  bad input  not JSON at all
  [500]   2.1s  bad input  JSON list, not dict
  [500]   2.1s  bad input  query is a number

Scores this trial:
  1. Every request leaves a complete log record    9/11
  2. Bad input gets a 400 with a message           3/5
  3. Matching query answers in under 8s            5/5
  4. One run costs at most 4000 tokens             5/5
  5. Original agent still works over HTTP          6/6
  successful runs: 5, median 2.63s, max 5.59s
  tokens per successful run: mean 714, min 655, max 847
```

Output capstone_trials.py – label before – trial 2

```
>> python capstone_trials.py --label before --trial 2
Trial 2 (before) → http://localhost:5000/ask

  [200]   4.3s  matching   vintage graphic tee under $30
  [200]   4.4s  matching   denim jacket under $50
  [200]   4.5s  matching   90s track jacket in size M
  [200]   4.8s  matching   silk slip dress in midi length under $40
  [200]   5.1s  matching   platform sneakers size 8
  [200]   2.7s  impossible designer ballgown size XXS under $5
  [400]   2.0s  bad input  empty JSON object
  [400]   2.1s  bad input  blank query
  [400]   2.1s  bad input  not JSON at all
  [500]   2.0s  bad input  JSON list, not dict
  [500]   2.1s  bad input  query is a number

Scores this trial:
  1. Every request leaves a complete log record    9/11
  2. Bad input gets a 400 with a message           3/5
  3. Matching query answers in under 8s            5/5
  4. One run costs at most 4000 tokens             5/5
  5. Original agent still works over HTTP          6/6
  successful runs: 5, median 2.43s, max 3.02s
  tokens per successful run: mean 668, min 640, max 700

Wrote results\capstone_before_trial2.md
```

Output capstone_trials.py – label before – trial 3

```
>> python capstone_trials.py --label before --trial 3
Trial 3 (before) → http://localhost:5000/ask

  [200]   4.8s  matching   vintage graphic tee under $30
  [200]   4.9s  matching   denim jacket under $50
  [200]   4.5s  matching   90s track jacket in size M
  [200]   5.6s  matching   silk slip dress in midi length under $40
  [200]   4.7s  matching   platform sneakers size 8
  [200]   2.8s  impossible designer ballgown size XXS under $5
  [400]   2.0s  bad input  empty JSON object
  [400]   2.0s  bad input  blank query
  [400]   2.0s  bad input  not JSON at all
  [500]   2.1s  bad input  JSON list, not dict
  [500]   2.0s  bad input  query is a number

Scores this trial:
  1. Every request leaves a complete log record    9/11
  2. Bad input gets a 400 with a message           3/5
  3. Matching query answers in under 8s            5/5
  4. One run costs at most 4000 tokens             5/5
  5. Original agent still works over HTTP          6/6
  successful runs: 5, median 2.73s, max 3.60s
  tokens per successful run: mean 721, min 639, max 793

Wrote results\capstone_before_trial3.md
```

serve.py:175-176

Change serve.py:175-176. Replace

```
payload = request.get json(silent=True) or {}
query = (payload.get("query") or "").strip()
```

with

```
payload = request.get_json(silent=True)
if not isinstance(payload, dict):
    payload = {}
query = payload.get("query")
```

```
 query = query.strip() if isinstance(query, str) else ""
```

Final result

```
# Capstone trial 2 — after
Run at 2026-11-12T19:15:22 against http://localhost:5000/ask
Produced by `capstone_trials.py::main`; log lines written by
`serve.py::log_request`.

## Scores

| Criterion | Result |
|---|---|
| 1. Every request leaves a complete log record | 11/11 |
| 2. Bad input gets a 400 with a message | 5/5 |
| 3. Matching query answers in under 8s | 4/5 |
| 4. One run costs at most 4000 tokens | 5/5 |
| 5. Original agent still works over HTTP | 6/6 |

- successful runs: 5, median 2.48s, max 16.23s
- tokens per successful run: mean 686, min 618, max 730
```

Update serve.py:175-176

Change serve.py:175-176. Replace

```
payload = request.get_json(silent=True) or {}
query = (payload.get("query") or "").strip()
```

with

```
payload = request.get_json(silent=True)
if not isinstance(payload, dict):
    payload = {}
query = payload.get("query")
query = query.strip() if isinstance(query, str) else ""
```

capstone_trials.py – label after – trial 1

```
# Capstone trial 1 — after

Run at 2026-11-12T19:14:28 against http://localhost:5000/ask

Produced by `capstone_trials.py::main`; log lines written by `serve.py::log_request`.

## Scores

| Criterion | Result |
|---|---|
| 1. Every request leaves a complete log record | 11/11 |
| 2. Bad input gets a 400 with a message | 5/5 |
| 3. Matching query answers in under 8s | 5/5 |
| 4. One run costs at most 4000 tokens | 5/5 |
| 5. Original agent still works over HTTP | 6/6 |

- successful runs: 5, median 2.60s, max 4.35s
- tokens per successful run: mean 687, min 634, max 756
```

Five targets, three trials, one fix

- C1 every request logged: 11 of 11
- C2 bad input gets 400: 5 of 5
- C3 answer under 8 s: 4 of 5
- C4 one run ≤ 4,000 tokens: 5 of 5
- C5 old agent still works: 6 of 6